

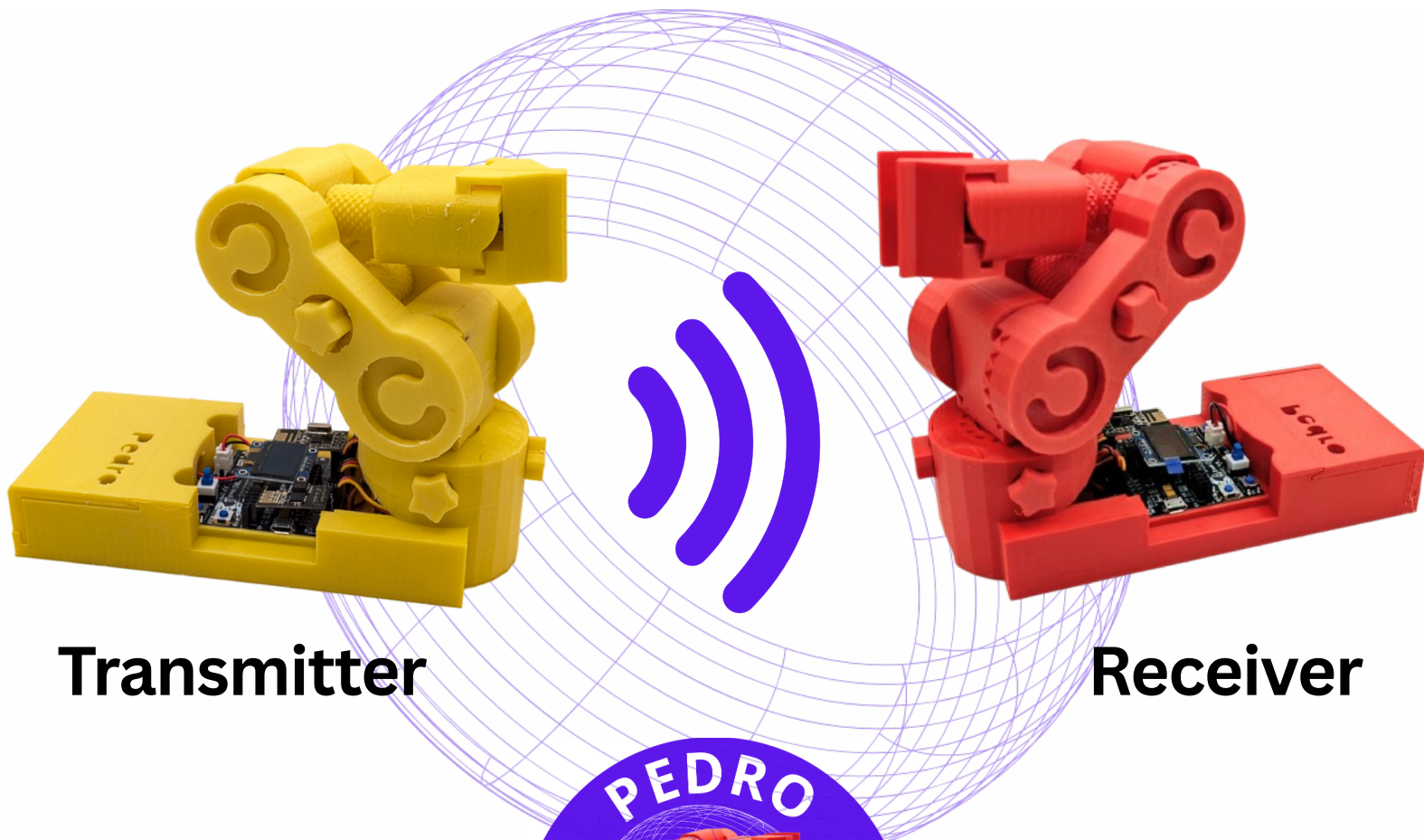
BEGINNER

pedro



STEM Program

Lesson #5: Radio Mode



Transmitter

Receiver

Complete this lesson
to earn your Pedro
Badge



Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Discover how robots can communicate wirelessly
using radio signals

Learning Goals

In this lesson, students will discover how robots can **communicate wirelessly** using radio signals.

By the end of the lesson, students will be able to:

- Understand how two robots exchange information
- Configure a transmitter and a receiver
- Control one robot remotely using another



At the end of this lesson, you will unlock your Pedro Badge.

the Pedro Manual Badge

Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 or 2 Assembled** Pedro robot **per student or per group**
(recommended: groups of 2 students)
 - **OLED Screen 128X64**
 - **Module NRF24L01**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



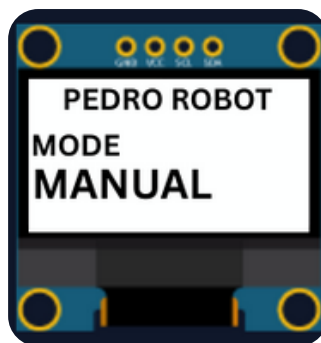
Board Configuration

(Skip this activity if the Pedro program is already uploaded)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions. Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
 - Tools → Manage Libraries → search **PedroRobot** → Install
 - Tools → Manage Libraries → search **U8glib** → Install
 - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✓ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.



Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo. The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

- LED D13 → servo D5 (Base)
- LED D11 → servo D6 (Shoulder)
- LED D8 → servo D9 (Elbow)
- LED D7 → servo D10 (Gripper)



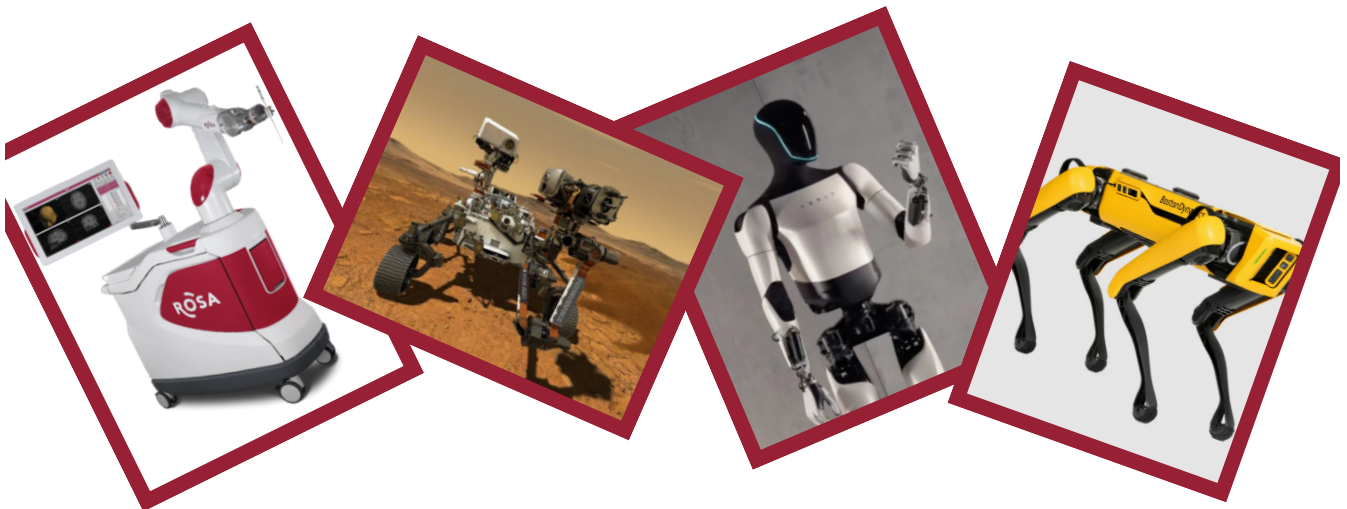
Remote Communication?

Remote communication in robotics allows robots and electronic devices to exchange information wirelessly in order to perform actions or complete tasks from a distance.

Wireless communication makes robotic systems more flexible because it eliminates the need for cables and allows robots to operate remotely, even in places that are difficult or dangerous for humans to reach.

Today, wireless communication is used in many fields, including:

- 🚀 **Space exploration** – controlling space rovers and satellites.
- 🚁 **Drones** – piloting unmanned aerial vehicles remotely.
- 🏥 **Medicine** – operating medical robots and monitoring patients.
- 🏠 **Smart homes (Home Automation)** – controlling connected devices such as lights, doors, and security systems.
- 🏭 **Industry** – coordinating autonomous robots in factories and warehouses.



Why is wireless communication important for robots that operate in dangerous or inaccessible environments?



What are the advantages and disadvantages of using wireless communication instead of cables in robotic systems?

What could happen if the communication signal between a robot and its controller is interrupted?



How Does Wireless Communication Work?

There are many different types of wireless communication. Each one uses a specific **communication protocol**, which is a set of rules that allows electronic devices to exchange information correctly.

Some of the most common wireless protocols are:

-  **Wi-Fi** – used to connect devices to the Internet.
-  **Bluetooth** – used to connect nearby devices such as headphones, keyboards, or robots.
-  **NRF24L01** – used for fast and reliable communication between electronic devices such as Pedro robots.

Although these technologies are different, they all work on the same principle: information is transmitted through **radio waves** at a specific **radio frequency**, measured in **Hertz (Hz)**.

Wireless communication takes place in two main steps:

- **Encoding (Transmission)**
Before sending a message, the transmitter **encodes** the information. Encoding means **converting instructions or data into a digital message** that can be transmitted by radio waves.
- **Decoding (Reception)**
The receiving robot captures the radio signal and **decodes** the message. Decoding means **reading and interpreting the received data** so the microcontroller understands what action to perform.

Radio Mode allows Pedro robots to communicate through radio signals using the NRF24L01 module. The robots can operate in two different roles:

- **Transmitter:** the Pedro robot that sends commands through radio signals.
- **Receiver:** the Pedro robot that receives and executes the commands.



How do robots know which commands to follow when they receive wireless signals?

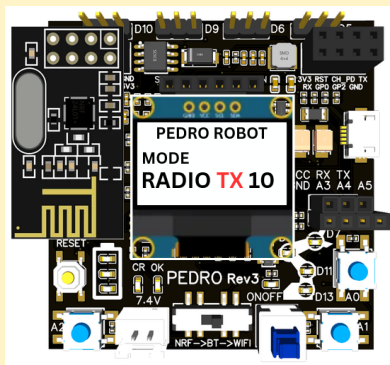
Design challenge: Imagine you are designing a robot for space exploration. What type of communication system would you choose and why?



Configure Pedro in Radio Mode

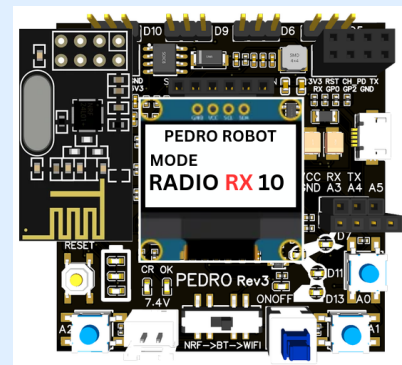
Configure Pedro as a Transmitter (TX)

1. Insert the **NRF24L01** modules.
2. Set Pedro's switch to **NRF (NRF → BT → WiFi)**
3. Restart Pedro
4. Enter to the Select Mode menu (hold **A0** for 4s)
5. Select **Radio mode**, then press and press **A0** to confirm.
6. Press **A1** to select the robot role: **TRANSMITTER**
7. Press **A0** to confirm the role.
8. Set the **Code Key** (the transmitter and receiver **must use the same Code Key**):
 - **A1** → increase the value
 - **A2** → decrease the value
 - Available values: **1 to 99**
9. Press **A0** to confirm the Code Key.
10. Press **A0** again to validate (**OK**).
11. Pedro is now configured as a **Transmitter (TX)** and is ready to send commands remotely. The screen should display the **Pedro role** and **Code Key (10 in this example)**.



Configure Pedro as a Receiver (RX)

1. Insert the **NRF24L01** modules.
2. Set Pedro's switch to **NRF (NRF → BT → WiFi)**
3. Restart Pedro
4. Enter to the Select Mode menu (hold **A0** for 4s)
5. Select **Radio mode**, then press and press **A0** to confirm.
6. Press **A1** to select the robot role: **RECEIVER**
7. Press **A0** to confirm the role.
8. Set the **Code Key** (the transmitter and receiver **must use the same Code Key**):
 - **A1** → increase the value
 - **A2** → decrease the value
 - Available values: **1 to 99**
9. Press **A0** to confirm the Code Key.
10. Press **A0** again to validate (**OK**).
11. Pedro is now configured as a **RECEIVER (RX)** and is ready to receive commands remotely. The screen should display the **Pedro role** and the **Code Key (10 in this example)**.



What happens when you select a servo on the Pedro Transmitter? The LED corresponding to the selected servo light on simultaneously on the Pedro Receiver. That indicates that the radio communication between both is established.

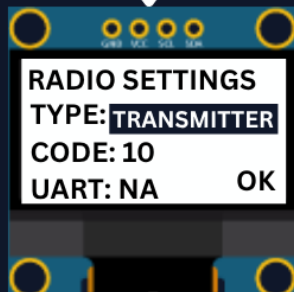
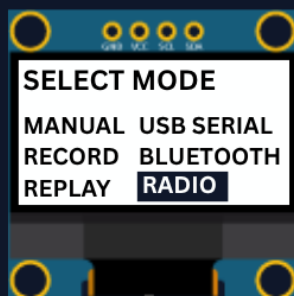
What happens when you move the Pedro robot Transmitter? The Pedro Receiver should follow the same movements simultaneously. Experiment with the radio communication by controlling all of Pedro's movements from the Transmitter.



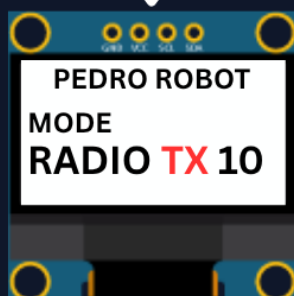
Radio Mode Configuration Overview

Pedro Radio Mode

Pedro Transmitter

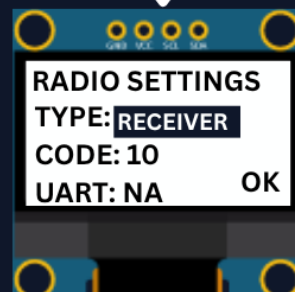
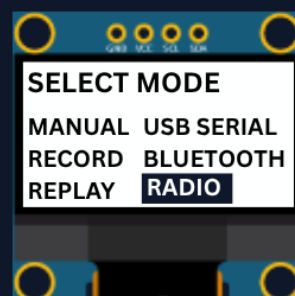


OK

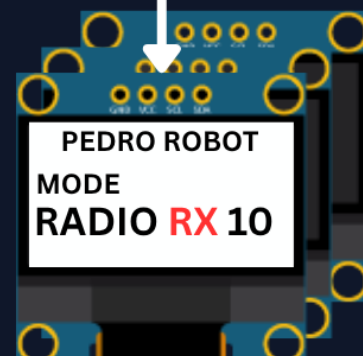


Enable Radio Mode
Set switch
NRF -> BT -> WiFi: to NRF

Pedro Receiver



OK

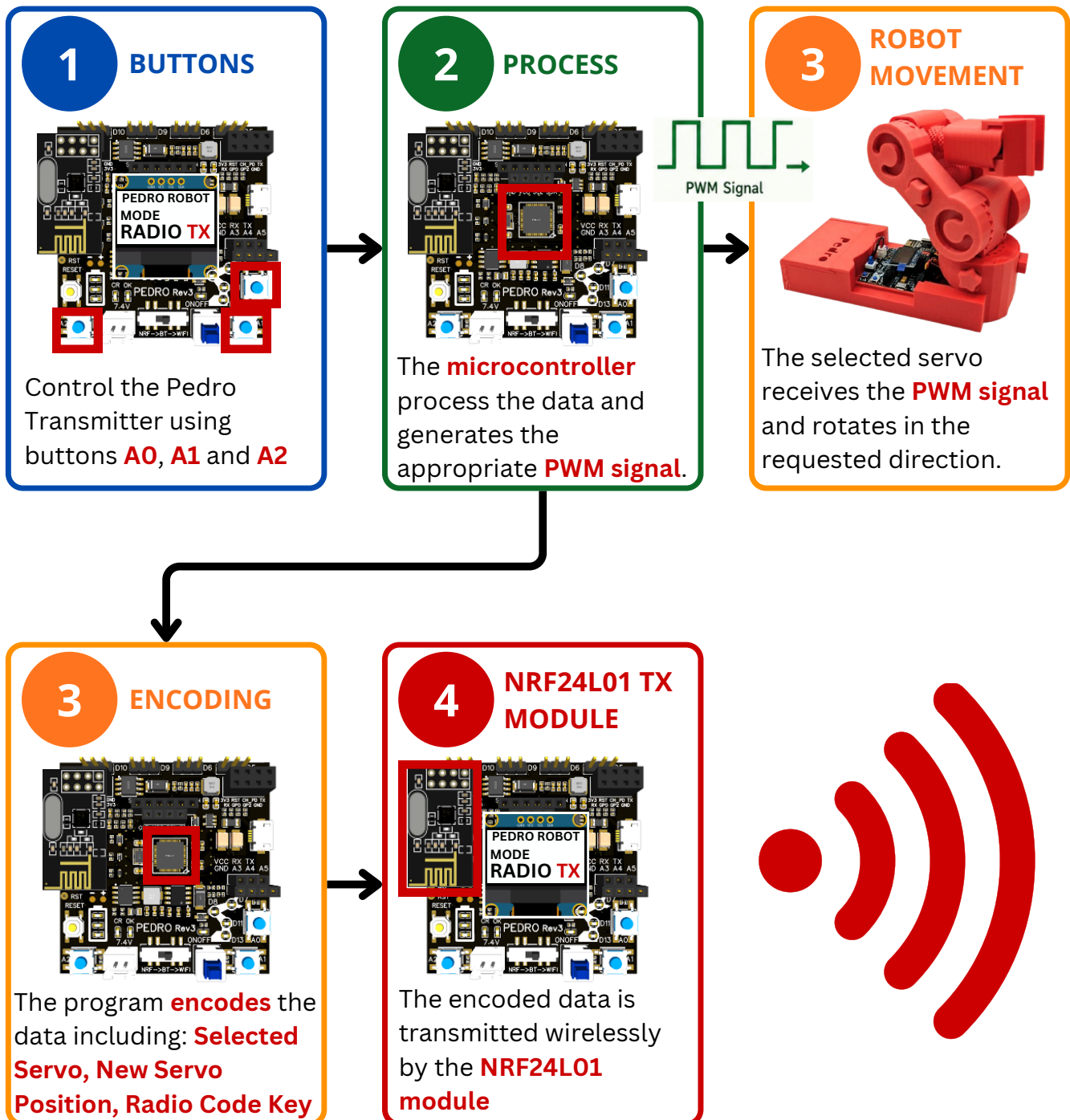


- Button A1/A2:
 - Select TRANSMITTER/RECEIVER
 - Set Code to 1 to 99
- Button A0: Validate





How Radio Transmission Works



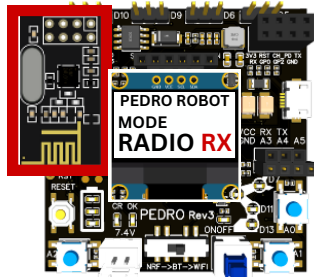


How Radio Reception Works



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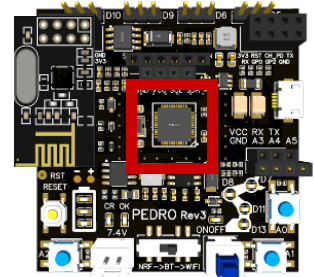
NRF24L01 RX MODULE



The **NRF24L01 Module** receives the data from the Transmitter.

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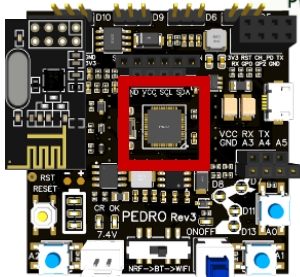
DECODING



The program **decodes** the data including: **Selected Servo, New Servo Position, Radio Code Key**

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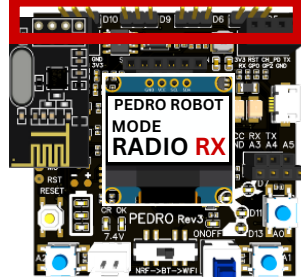
PROCESS



The **microcontroller** receives the data and generates the appropriate **PWM signal**.

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SERVO MOTORS



The selected servo receives the **PWM signal** and rotates in the requested direction.

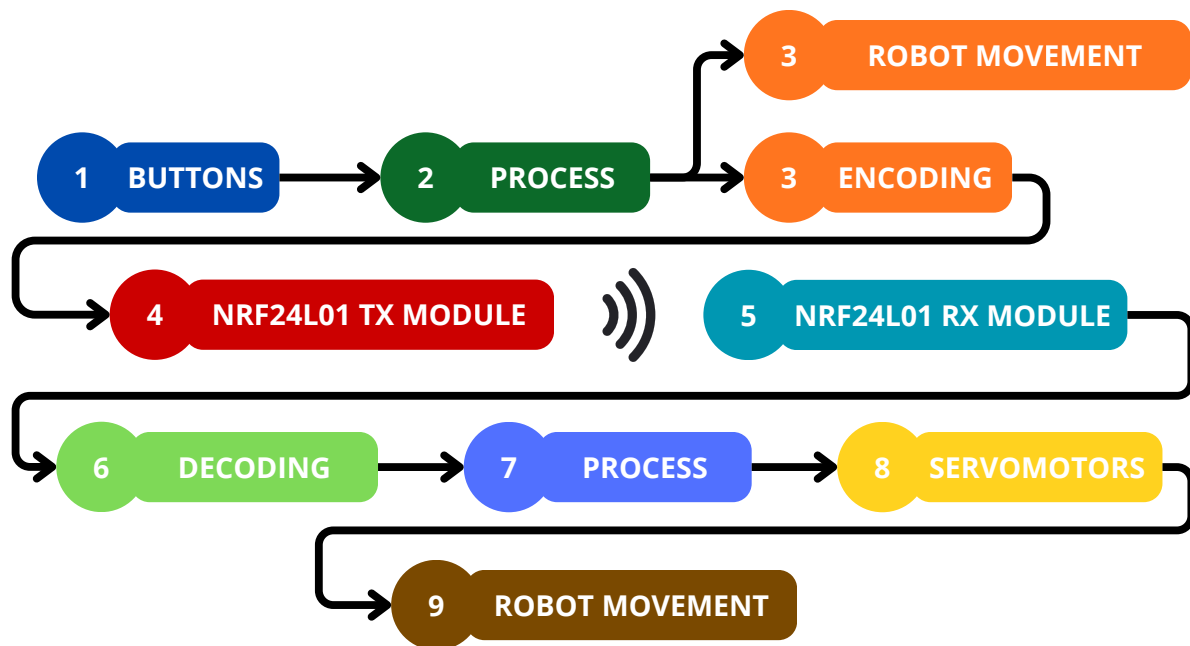
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REBOT MOVEMENT



The selected servo rotates, causing the robot to perform the requested movement.

The Radio communication chain is:



Activity 1 – Interrupted Communication

Student Instructions:

In this activity, you will investigate how the **radio code key** affects communication between a Pedro **Transmitter** and **Receiver**.

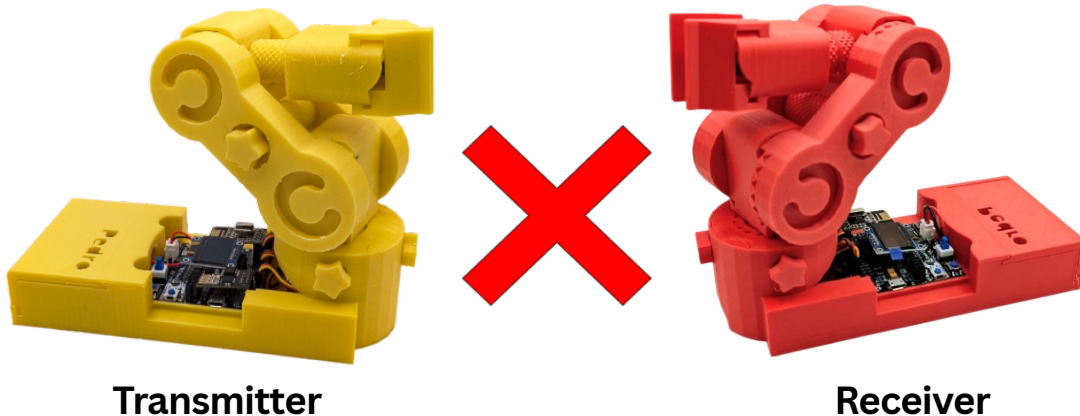
For two Pedro robots to communicate successfully, they must use the **same radio code key**. The code key acts like a **shared password** or **digital identifier** that allows the Receiver to recognize and accept messages from the correct Transmitter. If the code keys do not match, the Receiver ignores the incoming messages.

Through this experiment, you will discover how **encoding** helps ensure that only the intended Receiver responds to the transmitted commands. This principle is widely used in many wireless technologies, such as remote controls, smart devices, and robotic systems, to improve communication reliability and security.

Procedure:

1. Change the Transmitter code key so that it no longer matches the

- Receiver code key. (for example, increase the value by 1)
2. Move the Pedro Transmitter and observe the behavior of the Receiver.
 3. Explain why the Pedro Receiver no longer responds to the Transmitter. Use the concept of encoding to justify your answer.
 4. Now change the Receiver code key to match the Transmitter code key.
 5. Move the Pedro Transmitter and observe the Receiver's behavior.
 6. The communication should be restored, allowing the Pedro Receiver to respond and follow the Transmitter's movements again.



Activity 2 – Measuring the Communication Range

Student Instructions:

Students will measure the maximum communication distance between two Pedro robots. They will compare the real-world performance with the theoretical specifications of the NRF24L01 module.

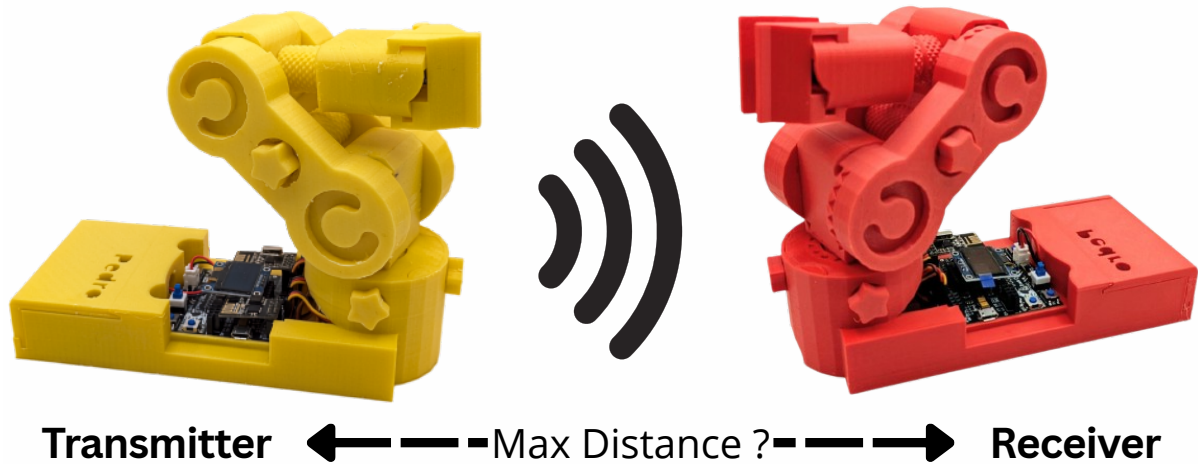
According to the manufacturer:

- **NRF24L01 typical range:** 30–50 meters indoors, up to 100 meters in open space.
- **Note:** Real performance depends on the environment, obstacles, interference, and power supply conditions.

Procedure:

1. Move the Pedro Receiver farther away from the Pedro Transmitter while

- the Transmitter is sending movement commands.
2. Observe when the Receiver stops responding to the Transmitter.
 3. Record the maximum distance at which communication is still working.
 4. Use this value to determine the communication range limit of the Pedro robots.



Activity 3 – Multiple Robot Communication

Student Instructions:

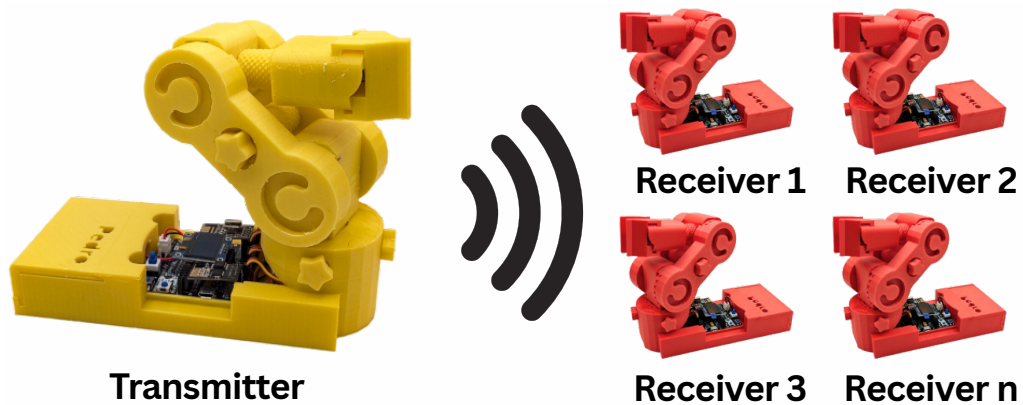
In this activity, you will explore how **one Pedro Transmitter** can control **multiple Pedro Receivers** simultaneously using radio communication.

When all Pedro Receivers are configured with the **same radio code key**, they recognize the commands sent by the Transmitter and respond at the same time. This demonstrates how a single wireless controller can coordinate the movements of several robots

Procedure:

1. Configure **one Pedro Transmitter** and **several Pedro Receivers** using the **same radio code key**.
2. On the Pedro Transmitter, **select the Base servo**.
3. Verify that the **Base servo LED** lights up simultaneously on all Pedro Receivers.
4. Move the Base servo on the Pedro Transmitter and observe the behavior of all the Receivers.

5. Experiment by controlling each of Pedro's servos from the Transmitter and observe how all the Receivers respond.



- Why do all the Pedro Receivers respond to the same commands?
- What role does the radio code key play in allowing multiple robots to communicate with one Transmitter?
- What advantages are there to controlling several robots at the same time?
- How could you modify the system so that only one Receiver responds while the others ignore the command?



Quiz - Check Your Understanding

1. Radio Mode enables communication between _____ robots.

- ☐ Two
- ☐ One
- ☐ Three
- ☐ Multiple

2. The NRF24L01 module is used for:

- ☐ Power
- ☐ Display
- ☐ Wireless communication
- ☐ Sound

3. A transmitter:

- ☐ Receives data
- ☐ Sends data
- ☐ Stores data
- ☐ Deletes data

4. A receiver:

- ☐ Sends data
- ☐ Receives data
- ☐ Prints data
- ☐ Ignores data

5. The Code Key is:

- ☐ Voltage
- ☐ Password for communication
- ☐ Speed
- ☐ Memory

6. If Code Keys are different:

- ☐ Works faster
- ☐ No communication
- ☐ LED brighter
- ☐ Nothing changes

7. The Code Key range is:

- ☐ 0-9
- ☐ 1-99
- ☐ 100-999
- ☐ 0-255

8. LED sync indicates:

- ☐ Error
- ☐ No battery
- ☐ Connection established
- ☐ Shutdown

9. Robots moving together is called:

- ☐ Isolation
- ☐ Synchronization
- ☐ Rotation
- ☐ Compilation

10. In Radio Mode, both robots must use the same:

- ☐ Battery level
- ☐ Serial port
- ☐ Code Key
- ☐ USB cable

Congratulations Mission Completed!

You have unlocked: 🏆 the Pedro Radio Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

